SQL - AGGREGATE FUNCTIONS & GROUPING

QUESTION 1
COUNT () Function

```
SELECT *
FROM employees;
```

OUTPUT

ID	first_name	last_name	department	Salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	55000	2020-08-10	San Francisco

Q2 ~~SELECT DISTINCT~~
Sum FUNCTION

```
SELECT Sum (salary) AS total_IT. salary
FROM employees
WHERE department = 'IT';
```

Table

total_IT_salary
220000

Q3 AVG() Function

```
SELECT AVG(salary) AS avg_hr_salary
FROM employees
WHERE department = 'HR';
```

OUTPUT

avg_hr_salary

49500,00

Q4 Min() & Max() function

```
SELECT MAX(salary) AS highest_salary, Min(salary)
AS lowest_salary
FROM employees;
```

output =

highest_salary	lowest_salary
62000	48000

Q5 GROUP BY Statement

```
SELECT department, Sum(salary) AS total_salary
FROM employees
GROUP BY department;
```

Output =

department	total_salary
Finance	119000
HR	99000
IT	220000
Marketing	105000

Q6 GROUP BY & COUNT()

```
SELECT department, COUNT(*) AS employee_count
FROM employees
GROUP BY city;
```

Output =

City	employee_count
Chicago	3
Houston	1
Los Angeles	2
New York	2
San Francisco	2

Q7 GROUP BY & ORDER BY

```
SELECT department, AVG(Salary) AS avg_Salary
FROM employees
GROUP BY department
ORDER BY avg_Salary DESC;
```

Output -

department	avg_Salary
IT	55 000
Finance	59 500
Marketing	52 500
HR	49500

Q8 HAVING CLAUSE

```
SELECT department, Sum (Salary) AS total_salary
FROM Employees
GROUP BY department
HAVING Sum (salary) > 100 000;
```

department	total_salary
Finance	119 000
IT	220 000

Q9 Combining GROUP BY, HAVING, and ORDER BY

```
SELECT city, COUNT(*) AS employee_count
FROM employees
GROUP BY city
HAVING COUNT(*) > 1
ORDER BY employee_count DESC;
```

City	employee_count
New York	2
Chicago	2
Los Angeles	2
San Francisco	2

Q10 Combining Aggregate Functions

```
SELECT department, AVG(salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC
LIMIT 1;
```

department	avg_salary
Finance	59 500